

Clear Away (CA) Normalisation

UNF

CONTRACT_WASTE(contract_num, waste_type_id, waste_type_desc, cw_collection_freq, (collrun_date, truck_vin, truck_rego, truck_make, truck_model, truck_year, (bin_rfid, bs_collected_kgs, bs_overweight)))

1NF

CONTRACT_WASTE(contract_num, waste_type_id, waste_type_desc, cw_collection_freq)

Candidate Keys:

- contract_num, waste_type_id

COLL_RUN (contract_num, waste_type_id, collrun_date, truck_vin, truck_rego, truck_make, truck_model, truck_year)

Candidate Keys:

- contract_num, waste_type_id, collrun_date

COLLECTION(contract_num, waste_type_id, collrun_date, bin_rfid, bs_collected_kgs, bs_overweight)

Candidate Keys:

- contract_num, waste_type_id, collrun_date, bin_rfid

Partial Dependency:

- waste_type_id $\rightarrow$ waste_type_desc
- collrun_date, bin_rfid $\rightarrow$ bs_collected_kgs, bs_overweight

2NF

CONTRACT_WASTE(contract_num, waste_type_id, cw_collection_freq)

COLL_RUN (contract_num, waste_type_id, collrun_date, truck_vin, truck_rego, truck_make, truck_model, truck_year)

COLLECTION(contract_num, waste_type_id, collrun_date, bin_rfid)

WASTE_TYPE(waste_type_id, waste_type_desc)

Tay Chee Hsian

BIN_STATUS(collrun_date, bin_rfid, bs_collected_kgs, bs_overweight)

Transitive Dependency:

- truck_vin $\rightarrow$ truck_rego, truck_make, truck_model, truck_year

3NF

CONTRACT_WASTE(contract_num, waste_type_id, cw_collection_freq)

COLL_RUN(contract_num, waste_type_id, collrun_date, truck_vin)

COLLECTION(contract_num, waste_type_id, collrun_date, bin_rfid)

WASTE_TYPE(waste_type_id, waste_type_desc)

BIN_STATUS(collrun_date, bin_rfid, bs_collected_kgs, bs_overweight)

TRUCK(truck_vin, truck_rego, truck_make, truck_model, truck_year)

Full Dependency:

- contract_num, waste_type_id $\rightarrow$ cw_collection_freq
- contract_num, waste_type_id, collrun_date $\rightarrow$ truck_vin
- waste_type_id $\rightarrow$ waste_type_desc
- collrun_date, bin_rfid $\rightarrow$ bs_collected_kgs, bs_overweight
- truck_vin $\rightarrow$ truck_rego, truck_make, truck_model, truck_year